UNIT 5 · MATHEMATICAL INTUITION BRIDGES

The Fokker-Planck Equation

Session 22 · How distributions of LP states evolve through time

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Primary Text: Liquidity Illusion (Forthcoming, 2026)

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What we'll cover today

1. **Forward Kolmogorov / Fokker-Planck**
   Distribution dynamics
2. **Derivation intuition**
   Conservation of probability
3. **Stationary solutions**
   Long-run distribution
4. **GE-LAV application**
   Cross-sectional LP state distribution
5. **Numerical methods preview**
   Finite-difference solvers

Fokker-Planck: how probability flows

For SDE $dL = \mu(L,t)dt + \sigma(L,t)dW$, the density $p(L,t)$ satisfies:

$$\frac{\partial p}{\partial t} = -\frac{\partial}{\partial L}[\mu(L,t)\,p] + \frac{1}{2}\frac{\partial^2}{\partial L^2}[\sigma^2(L,t)\,p]$$

Two terms with physical meaning:

Advection: $-\partial/\partial L\ [\mu(L,t)\cdot p]$

How the drift pulls probability mass along the L axis.

Diffusion: $\frac{1}{2}\cdot\partial^2/\partial L^2\ [\sigma^2(L,t)\cdot p]$

How random shocks spread out the probability mass.

Session 22 summary

What we accomplished today

1. Fokker-Planck (forward Kolmogorov) describes how SDE densities evolve through time
2. Two terms: advection (drift) + diffusion (noise spreading)
3. OU process has explicit stationary solution: Normal($\bar{L}$, $\sigma^2/2\kappa$)
4. GE-LAV uses Fokker-Planck for the cross-sectional LP distribution evolution

Next session

Session 23: Jensen's inequality and convexity — the math behind Failure 2